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1. Prove Fermat's Little Theorem from group-theoretic first principles, then explain how it motivates the Miller–Rabin probabilistic primality test. Apply the test (base $a = 2$) to $n = 561$ and explain why 561 is a Carmichael number that fools the simpler Fermat test. (6 marks)
2. Prove that Z_n , under addition and multiplication modulo n , is a field if and only if n is prime. Illustrate your proof by contrasting the multiplicative structure of Z_{11} (prime modulus) with Z_{12} (composite modulus). (5 marks)
3. Construct $GF(2^3)$ using the irreducible polynomial $x^3 + x + 1$ over $GF(2)$. Build the multiplication table for a generator element α , and determine whether the multiplicative group $GF(2^3)^*$ is cyclic. (6 marks)
4. Derive the time complexity of modular exponentiation via square-and-multiply. Compute $7^{560} \bmod 561$ step by step, and interpret the result in light of 561 being a Carmichael number. (5 marks)
5. Compare deterministic (AKS) and probabilistic (Miller–Rabin) primality testing in terms of asymptotic complexity, error probability, and practical relevance to RSA key generation at 2048-bit and 4096-bit key sizes. (4 marks)
6. Derive the Extended Euclidean Algorithm formally (as a sequence of linear combinations), then use it to compute $\gcd(240, 46)$ together with the Bézout coefficients x, y such that $240x + 46y = \gcd(240, 46)$. (5 marks)
7. State and prove the Chinese Remainder Theorem (existence and uniqueness) for pairwise coprime moduli. Apply it to solve: $x \equiv 2 \pmod{3}$, $x \equiv 3 \pmod{5}$, $x \equiv 4 \pmod{7}$, $x \equiv 5 \pmod{11}$ (6 marks)

8. Derive the CRT-based RSA decryption/signing optimization from first principles (computing d_P , d_Q , and q_{inv}), and demonstrate the $\sim 4\times$ speed-up with a fully worked numerical example using small primes p, q . (6 marks)
9. Explain the Bellcore fault-injection attack on RSA-CRT signatures: how a single faulty modular computation during signing leaks a factor of n , and what countermeasures (e.g., signature verification before release) prevent it. (5 marks)
10. Compute a modular inverse using both the Extended Euclidean Algorithm and Fermat's-Little-Theorem-based exponentiation (for a prime modulus), and discuss the trade-offs between the two approaches for large moduli. (4 marks)
11. Design and fully execute a Playfair cipher encryption and decryption of a phrase of your choosing (minimum 16 letters) with a keyword of your choosing. Present the complete 5×5 key matrix as a table, and show every digraph substitution step. (6 marks)
12. Critically analyze the structural weaknesses of the Playfair cipher against digraph frequency analysis, and explain why it remains vulnerable to known-plaintext attacks despite being far stronger than monoalphabetic substitution. (5 marks)
13. Derive and trace **two full rounds** of a simplified Feistel-structured cipher (DES-like), showing $L_0, R_0 \rightarrow L_1, R_1 \rightarrow L_2, R_2$ symbolically, then with the numeric values $L_0 = 0xAAAAAAAA$, $R_0 = 0xF0F0F0F0$, $K_1 = 0x0F0F0F0F$, $K_2 = 0x33333333$, using XOR as the round function. (6 marks)
14. Formally define the *avalanche effect* in block ciphers. Using a toy 8-bit substitution-permutation network of your own construction, demonstrate how flipping a single input bit changes at least half the output bits, and explain why this property is essential for cryptographic security. (5 marks)
15. Evaluate the brute-force resistance of an 8-character alphanumeric password (62-character alphabet) versus a 6-digit PIN, under a modern GPU cracking rate of 10^9 guesses/second. Compute worst-case and expected-case cracking times for both, and discuss the implications for password policy design. (4 marks)
16. Compare the error-propagation and parallelizability characteristics of ECB, CBC, CFB, OFB, and CTR modes using a labelled diagram for each. Recommend the most suitable mode for (a) encrypting a live video stream over a lossy network and (b) encrypting a static disk image, with justification. (6 marks)

Mode	Parallel encryption	Parallel decryption	Error propagation on bit-flip	Requires unique IV/nonce
ECB	Yes	Yes	Single block only (but reveals patterns)	No
CBC	No	Yes	Corrupts current + next block	Yes
CFB	No	Yes	Corrupts current + next block	Yes
OFB	No	No	Single bit only	Yes (never reused)
CTR	Yes	Yes	Single bit only	Yes (never reused)

17. Using the partial AES S-box below, perform the SubBytes transformation on the state word $[0x23, 0xA7, 0x4C, 0x19]$, then apply ShiftRows to a 4×4 state matrix that has this word as its first row (fill remaining rows with $0x00$ for the purpose of the exercise). Report the resulting matrix. (6 marks)

Row ↓ / Col →	3	4	7	9	A	C
1	6D			C6		
2	D4					
4		A1				2E
A			63		D2	

18. Derive the AES-128 Rijndael key schedule and compute the first **two** round keys (K_1, K_2) from a given 128-bit cipher key of your choosing, showing the RotWord, SubWord, and Rcon steps explicitly. (6 marks)

- 19.** Explain the MixColumns transformation as polynomial multiplication over $GF(2^8)$ modulo $x^8 + x^4 + x^3 + x + 1$. Work through the full multiplication of one input column by the fixed MixColumns matrix, showing the $GF(2^8)$ arithmetic for each byte. (5 marks)
- 20.** Critically evaluate AES's resistance to side-channel attacks, specifically timing analysis (e.g., cache-timing attacks on table-based S-box implementations) and differential power analysis. Describe at least two concrete countermeasures (e.g., constant-time S-box, masking) and their performance trade-offs. (4 marks)
- 21.** Derive the RSA key-generation, encryption, and decryption procedures from Euler's Theorem, and prove correctness of decryption (i.e., that $(M^e)^d \equiv M \pmod{n}$) for the general case, including the case $\gcd(M, n) \neq 1$. (6 marks)
- 22.** Given primes $p = 11$, $q = 13$, generate a full RSA keypair (choosing a valid public exponent e), then sign a message hash $H(M) = 23$ and verify the signature, showing all intermediate modular exponentiation steps. (6 marks)
- 23.** Derive a general formula for IPSec ESP overhead percentage in tunnel mode as a function of payload size, header size, authentication-data size, and padding, then evaluate it for a 1200-byte payload with an 8-byte ESP header, 16 bytes of authentication data, and 2 bytes of padding. Discuss how this overhead scales for small IoT-sized packets versus large payloads. (5 marks)
- 24.** Execute a full Diffie–Hellman key exchange with $p = 23$, $g = 5$, private keys $a = 6$ (Aleya) and $b = 15$ (Badol), confirming both parties derive the same shared secret. Then explain the man-in-the-middle vulnerability of unauthenticated DH and describe how an authenticated variant (e.g., the Station-to-Station protocol) mitigates it. (6 marks)
- 25.** Compare the computational cost of establishing a shared key via RSA key transport versus Diffie–Hellman key agreement at an equivalent security level (e.g., 3072-bit RSA/DH versus 256-bit ECDH). Discuss why forward secrecy favors (EC)DHE over RSA key transport in modern TLS. (4 marks)
- 26.** In an ElGamal scheme with $p = 467$, $g = 2$, private key $x = 127$, message hash $H(M) = 100$, and nonce $k = 213$ (with $\gcd(k, p - 1) = 1$): compute the public key, generate the signature (r, s) , and verify it. (6 marks)
- 27.** Prove algebraically that the ElGamal signature verification equation $g^{H(M)} \equiv y^r r^s \pmod{p}$ holds for a correctly generated signature, starting from the definition $s \equiv k^{-1}(H(M) - xr) \pmod{p - 1}$. (5 marks)

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Client AS TGS Service (S)

|--(1) ID C, ID TGS----->|

```

```

|<-(2) {K_c,tgs}Kc, TGT-----|
|
|--(3) Authenticator, TGT----->|
|
|<-(4) {K_c,s}Kc,tgs, Ticket_S-----|
|
|--(5) Authenticator, Ticket_S----->|
|
|<-(6) {timestamp+1}Kc,s-----|

```

**37.** Explain why a plain hash-based MAC (i.e.,  $H(K \parallel M)$ ) is vulnerable to a length-extension attack, and derive how HMAC's nested construction

$H(K \oplus o, p, adH(K \oplus i, p, adM))$  eliminates this vulnerability. Illustrate with a small worked (simplified numeric) MAC example. (5 marks)

**38.** Diagram the TLS 1.3 handshake (ClientHello/ServerHello with key shares, encrypted extensions, Finished messages), and explain how the session keys are derived via HKDF from the (EC)DHE shared secret. Compare this against the TLS 1.2 handshake in terms of round-trip count and forward secrecy. (6 marks)

**39.** Design a bandwidth and duty-cycle budget for an IoT sensor network in which each node transmits a 1200-byte packet once per minute over a LPWAN technology of your choosing (e.g., LoRaWAN). Compute the total data volume per node per day (in bytes and KB), and discuss regulatory duty-cycle constraints (e.g., 1% airtime limits) that could affect this design. (5 marks)

**40.** Critically assess a lightweight cryptographic standard suitable for resource-constrained IoT devices (e.g., NIST's Ascon, selected as the lightweight cryptography standard) against AES-128 in terms of memory footprint, energy consumption, and side-channel resistance. Recommend which is more appropriate for a battery-powered sensor with < 10 KB of RAM. (4 marks)